



Ahsanullah University of Science & Technology

Department of Computer Science & Engineering

Course No : CSE 2214
Course Title : Assembly Language Programming Sessional
Assignment No 2

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Question No: 01

Question: Write a program that initializes a variable health with an initial value of 100. Take a choice and a factor value (between 0 to 9) as input from the user.

- If the user enters H, increase health by 2 * factor
- If the user enters D, decrease health by 2 * factor.
- If the user enters S, keep the score unchanged. No need to display the final health value.

Answer:

```
.MODEL SMALL
.STACK 100H

.DATA
HEALTH DB 100
CHOICE DB ?
FACTOR DB ?

.CODE
MAIN PROC

    MOV AX,@DATA
    MOV DS,AX

    ; Read choice
    MOV AH,1
    INT 21H
    MOV CHOICE,AL

    ; Read factor
    MOV AH,1
    INT 21H
    SUB AL,30H
    MOV FACTOR,AL

    CMP CHOICE,'S'
    JE EXIT

    CMP CHOICE,'H'
    JE INCREASE

    CMP CHOICE,'D'
    JE DECREASE

    JMP EXIT

INCREASE:
    MOV BL,FACTOR
```

```

SHL BL,1
ADD HEALTH,BL
JMP EXIT

DECREASE:
MOV BL,FACTOR
SHL BL,1
SUB HEALTH,BL

EXIT:
MOV AH,4CH
INT 21H

MAIN ENDP
END MAIN

```

Question No: 02

Question: Write a program that reads characters until ENTER is pressed and counts the number of vowels and special characters. Assume your inputs contain only lowercase letters, digits and special characters.

Answer:

```

.MODEL SMALL
.STACK 100H

.DATA
MSG1 DB '$'
MSG2 DB 0DH,0AH,'VOWELS: $'
MSG3 DB 0DH,0AH,'SPECIAL CHARACTERS: $'

VOWELS DB 0
SPECIAL DB 0

.CODE
MAIN PROC

    MOV AX,@DATA
    MOV DS,AX

    MOV AH,9
    LEA DX,MSG1
    INT 21H

AGAIN:

    MOV AH,1
    INT 21H

    CMP AL,0DH

```

```

JE PRINT

CMP AL,'a'
JE COUNT_VOWEL
CMP AL,'e'
JE COUNT_VOWEL
CMP AL,'i'
JE COUNT_VOWEL
CMP AL,'o'
JE COUNT_VOWEL
CMP AL,'u'
JE COUNT_VOWEL

CMP AL,'0'
JB COUNT_SPECIAL

CMP AL,'9'
JBE AGAIN

CMP AL,'a'
JB COUNT_SPECIAL

CMP AL,'z'
JBE AGAIN

COUNT_SPECIAL:
INC SPECIAL
JMP AGAIN

COUNT_VOWEL:
INC VOWELS
JMP AGAIN

PRINT:

MOV AH,9
LEA DX,MSG2
INT 21H

MOV DL,VOWELS
ADD DL,30H
MOV AH,2
INT 21H

MOV AH,9
LEA DX,MSG3
INT 21H

MOV DL,SPECIAL
ADD DL,30H
MOV AH,2

```

INT 21H

MOV AH,4CH
INT 21H

MAIN ENDP
END MAIN